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(71)Name of Applicant :
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India - 639113 Karur -----
Name of Applicant : NA
Address of Applicant : NA

(72)Name of Inventor :
1)Dr. P.JEYAKUMAR
Address of Applicant :Assistant Professor, Department of ECE, M.Kumarasamy College of Engineering, Karur, India-639113 Karur -----
2)Dr.S.JEGADEESAN
Address of Applicant :Professor, Department of ECE, M.Kumarasamy College of Engineering, Karur, India-639113 Karur -----
3)S.VISHWA
Address of Applicant :UG Student, Department of ECE, M.Kumarasamy College of Engineering, Karur, India-639113 Karur -----
4)P.SUBASH
Address of Applicant :UG Student, Department of ECE, M.Kumarasamy College of Engineering, Karur, India-639113 Karur -----
5)B.SANJAI
Address of Applicant :UG Student, Department of ECE, M.Kumarasamy College of Engineering, Karur, India-639113 Karur -----
6)V.PRADEEP
Address of Applicant :UG Student, Department of ECE, M.Kumarasamy College of Engineering, Karur, India-639113 Karur -----
7)R.K.SHANJAY
Address of Applicant :UG Student, Department of ECE, M.Kumarasamy College of Engineering, Karur, India-639113 Karur -----
8)M.SIVA RAMAKRISHNAN
Address of Applicant :UG Student, Department of ECE, M.Kumarasamy College of Engineering, Karur, India-639113 Karur -----
9)S.SUBASH
Address of Applicant :UG Student, Department of ECE, M.Kumarasamy College of Engineering, Karur, India-639113 Karur -----

(57) Abstract :
ABSTRACT A SYSTEM FOR PRESERVING TOMATOES USING EVAPORATIVE COOLING The present invention discloses a system (20) for preserving the tomatoes using evaporative cooling method for increasing the shelf life of the tomatoes. The preservation system (20) comprises a cooler housing, papaya fibre reinforced tray (5), solar panels (8) and cooling pad (11). The cooler housing (3) is maintaining minimum temperature with the integration of climate control mechanism (9) for escaping moist air and allowing fresh air inside the preservation system. The plurality of papaya fibre reinforced tray (5) is configured to pack and preserve tomatoes inside the system for increasing the lifetime by means of moisture resistant coatings. The solar panels (8) located at the top of the preservation system which is configured to generate the electricity from the sunlight and store it in a battery for continuous supply of power to the system. The cooling pad (11) is configured to provide the cooling nature and airflow to the preservation system. The cooling pad (11) comprises a porous jute bag equipped with frame mesh which is configured to absorb the water while dispensing from the water return pump (10) and to supply the cooling air by the inward rotation of propeller fan (13) for controlling the temperature. The present invention (20) will increase the lifetime of the tomatoes using evaporative cooling in an effective manner. Fig 1

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